

PL/SQL Beginner Programs & Loops

Gate Smashers Lesson Revision Notes • Syntax, Logic & Control Structures

QUICK REFERENCE

1. BASIC PL/SQL BLOCK STRUCTURE

A standard PL/SQL block consists of mandatory and optional declarative/executable sections:

```
DECLARE
  -- Declarative section (variables)
BEGIN
  -- Executable section (logic)
END;
/
```

★ Key Rules

- **DECLARE** → Declare variables
- **BEGIN** → Start block execution
- **END;** → Terminate the PL/SQL block

Declarative Section

Used to define variables and data types prior to execution.

```
a INT;
b INT;
c INT;
```

Executable Section

Contains logic, computations, and database queries placed between **BEGIN** and **END;** / .

2. OPERATORS & INPUT HANDLING

Assignment Operator

In PL/SQL, use **:=** for assignment.

```
a := 10;
b := 20;
c := a + b;
```

⚠ **Distinction:** **:=** (Assignment) vs **=** (Comparison)

Taking User Input

Use substitution variables like **&a** and **&b** .

```
a := &a;
b := &b;
```

Note: In environments like Oracle Live SQL where substitution prompts aren't supported, assign direct values instead.

3. PROGRAM 1 — SUM OF TWO NUMBERS

Logic Flow: Input A, B → Compute **C = A + B** → Output C

PL/SQL Program

```
DECLARE
  a INT;
  b INT;
  c INT;
BEGIN
  a := 10;
  b := 20;

  c := a + b;

  DBMS_OUTPUT.PUT_LINE('Sum of a and b
= ' || c);
```

Output

Sum of a and b = 30

Output Handling (**DBMS_OUTPUT.PUT_LINE**)

Used to print lines to the console.

- Text: **DBMS_OUTPUT.PUT_LINE('Hello');**
- Variable: **DBMS_OUTPUT.PUT_LINE(c);**
- Concatenation: **DBMS_OUTPUT.PUT_LINE('Sum = ' || c);**

🧠 **Memory Trick:** **||** joins strings; **PUT_LINE** prints a line.

4. PROGRAM 2 — GREATEST OF TWO NUMBERS

Introduces conditional decision-making using the `IF...ELSE` construct.

PL/SQL Program

```
DECLARE
  a INT;
  b INT;
BEGIN
  a := 10;
  b := 20;

  IF a > b THEN
    DBMS_OUTPUT.PUT_LINE('A is
greater');
  ELSE
    DBMS_OUTPUT.PUT_LINE('B is
greater');
  END IF;
END;
/
```

Output

B is greater

Reasoning: Evaluates $10 > 20$ (False), executing the `ELSE` branch.

Printing Values with Text

```
DBMS_OUTPUT.PUT_LINE('B is greater and
value of B = ' || b);
-- Output: B is greater and value of B =
20
```

⚠ Common Mistake: Ending Statements

Do not confuse statement endings! `END IF;` closes a conditional block, while `END;` closes the main PL/SQL block.

🔥 IMPORTANT VIVA QUESTIONS

Q1. What is PL/SQL?

PL/SQL is Oracle's procedural extension to SQL that allows procedural structures like variables, conditions, and loops alongside SQL statements.

Q2. What is the declarative section?

The section starting with `DECLARE` where variables and constants are defined before execution.

Q3. What is the executable section?

The section contained between `BEGIN` and `END;` where procedural logic and queries execute.

Q4. What is the assignment operator in PL/SQL?

The `:=` operator.

Q5. How do you print output and end conditional/main blocks?

Use `DBMS_OUTPUT.PUT_LINE()` for output, `END IF;` for IF blocks, and `END;` for the PL/SQL block.

5. PL/SQL FOR LOOP

A FOR loop is used when the exact iteration range is known beforehand. It automatically handles loop variable initialization and incrementing.

Syntax & Structure

```
FOR variable IN lower .. upper
LOOP
```

Example Code

```
DECLARE
  a NUMBER;
```

